# Zhenglin Geng

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## Education

- **School of Software, Tsinghua University, Beijing** Aug 2012 - Present

**B.S. in Computer Science**, Major GPA: **93/100** (Ranking: **3/57**)

Core Courses: Computer System Software(1) (**97, top 1**), Computer System Software(2) (**99, top 4**), Data Structure and Algorithm(2) (**95, top 1**), Software Theory (**99, top 2**), Web Design and Produce (**100**), Web Front-end Technology (**100**), Probability and Statistics(**100**), Numerical Analysis (**100**), Physics for Scientists and Engineers B (**100**),

Awards: 2015 Meritorious Winner in Mathematical Contest in Modeling (MCM)
2015 National Endeavor Fellowship
2014 National Fellowship
2013 National Endeavor Fellowship

## Research Experience

- **Video Cropping Taking Aesthetic Factors into Account(Internship, MSRA)** Oct 2015 - present

Advisor: **Steve Lin**, Senior reseacher at Microsoft Research Asia

◊ Video cropping is different from image cropping because motion may changes the composition of the frame. I am designing experiment to take these aesthetic factors into account and generate visually pleasing videos.

- **Real-Time Rigid Body Simulation with Game Engine Integration (UGVR, Stanford)** Jun 2015 - Aug 2015

Advisor: **Ron Fedkiw**, Professor at department of computer science

◊ I built the physics simulation library (*PhysBAM, 3000+ files, 3.5m+ lines*) to a native plugin for a game engine (*Unity*) to achieve higher rendering performance.

◊ I developed a real-time rigid body simulation game including components like explosion, articulated bridge, falling rocks, reflecting mirror, cloth, fluid, trees and so on.

- **iFace——a one-click tool for face replacement** Jan 2014-May 2014

Advisor: **Bin Wang**, Associate Professor at School of Software, Tsinghua University

◊ We developed an android application, *iFace*, which identifies the human faces in pictures and allow users to replace them with faces taken from other pictures.

◊ We used *openCV* to identify the faces and *cocos2d* to run our algorithm on Android platform.

◊ Our app won the **Second Prize of Challenge Cup** in Tsinghua University in 2014.

- **Content Based Image Retrieval** May 2014 – May 2014

Advisor: **Xiaoming Jin**, Associate Professor at School of Software, Tsinghua University

◊ I realized the R-tree structure with advanced split-node method and different relevance ranking methods, and evaluated the performance of them during the image searching process.

◊ I compared different machine learning methods, like KNN, K-means, Naïve Bayesian, K-D tree, BP neural network and radial basis neural network in image retrieval process after the feature extraction procedure.

## Miscellanea

- **Programming skills**

◊ Familiar with C/C++, C#, Java, Python, Javascript, Matlab, Haskell.

◊ Familiar with Android development and web development.

◊ Familiar with gcc, gdb, CMake; Familiar with git.

◊ Familiar OpenCV, OpenGL, DirectX 11.

- **English skills**

◊ **TOEFL iBT** : 110 (Reading: 30, Listening: 30, Speaking: 23, Writing: 27).

◊ **GRE**: Verbal: 163, Quantitative: 170, Writing: 4.0.